🧾 Flask Project Structure – Complete Notes

🔹 1. What is Flask?

Flask is a **lightweight Python web framework** used to build web applications.

👉 You can create:

- Websites
- APIs
- Backend systems

🔹 2. Basic Flask Project Structure

```
flask_project_name/
│
├── app.py
├── templates/
│   ├── home.html
│   ├── contact.html
│   └── about.html
│
├── static/
│   ├── css/
│   │   └── style.css
│   ├── images/
│   │   ├── kajal.jpeg
│   │   └── rashmika.jpg
│   └── js/
│       └── home.js
│
└── requirements.txt
```

🔹 3. Explanation of Each Folder/File

✅ 1. `app.py` (Main File)

- This is the **entry point** of your Flask app
- Contains:
  - App creation
  - Routes (URLs)
  - Logic

👉 Example:

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
  return render_template("home.html")

if __name__ == "__main__":
  app.run(debug=True)
```

✅ 2. `templates/` Folder

- Stores **HTML files**
- Flask automatically looks here for UI pages

👉 Example files:

- `home.html`
- `contact.html`
- `about.html`

👉 Usage:

```
return render_template("home.html")
```

✅ 3. `static/` Folder

- Stores **frontend assets**

Inside static:

- `css/` → Stylesheets
- `images/` → Images
- `js/` → JavaScript files

👉 Example:

<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">

✅ 4. `requirements.txt` (Optional but Important)

- Stores project dependencies

👉 Example:

flask==2.3.0

👉 Install using:

pip install -r requirements.txt

🔹 4. Installing Flask

👉 First install Flask:

pip install flask

🔹 5. Why Virtual Environment (VERY IMPORTANT – Interview)

❓ Problem:

Different projects may need different Flask versions

👉 Example:

- Project A → Flask 2.3
- Project B → Flask 2.2

⚠️ If installed globally → conflict happens

✅ Solution: Virtual Environment (venv)

Each project gets its own environment

🔹 6. How to Create Virtual Environment

python -m venv venv

👉 This creates a folder:

venv/

🔹 7. How to Activate Virtual Environment

👉 Windows:

venv\Scripts\activate

👉 Mac/Linux:

source venv/bin/activate

👉 After activation:

(venv) C:\project>

🔹 8. Install Flask Inside venv

pip install flask

🔹 9. Full Setup Steps (Important for Interviews)

1. Create project folder
2. Open terminal inside folder
3. Create virtual environment

    python -m venv venv

4. Activate it

    venv\Scripts\activate

5. Install Flask

    pip install flask

6. Create `app.py`
7. Create `templates/` and `static/`

🔹 10. Real-Time Understanding

Without venv ❌

- All projects share same Flask
- Version conflicts happen

With venv ✅

- Each project is independent
- Safe and professional

🔹 11. Interview Questions (Must Know)

❓ What is Flask?

👉 Lightweight Python web framework

❓ What is `templates` folder?

👉 Stores HTML files

❓ What is `static` folder?

👉 Stores CSS, JS, Images

❓ Why use virtual environment?

👉 To avoid dependency conflicts

❓ What is `app.py`?

👉 Main file to run Flask app

🔹 12. Simple Summary

👉 Flask Project = 4 main parts:

- `app.py` → backend logic
- `templates/` → HTML
- `static/` → CSS, JS, Images

- `venv/` → isolated environment